

An Augmentation-Dimension Answer to a Question of Ferenczi–Galego

FA Banach run `fa_banach.001`, lane 16

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Status

This packet gives a likely valid full negative answer to Question 13 in Ferenczi–Galego, *Countable groups of isometries on Banach spaces* [1]. The question asks whether there are arbitrarily large finite groups G of order n for which $\{-1, 1\} \times G$ can be represented in a separable real Banach space X if and only if $\dim X \geq n$. The answer is no: for every finite group G with $|G| = n \geq 3$, the same group is already representable in every separable real Banach space of dimension at least $n - 1$.

Source Question

The source question appears as Question 13 on page 18 of [1].

Question 13 For arbitrarily large $n \in \mathbb{N}$, does there exist a group G with $|G| = n$, such that $\{-1, 1\} \times G$ is representable in a separable real Banach space X if and only if $\dim X \geq n$?

In transcription:

For arbitrarily large $n \in \mathbb{N}$, does there exist a group G with $|G| = n$, such that $\{-1, 1\} \times G$ is representable in a separable real Banach space X if and only if $\dim X \geq n$?

Here “representable” is used in the sense of [1]: a group H is representable in X if H is isomorphic to the full group of surjective linear isometries of X for some equivalent norm on X .

Proof Intuition

Ferenczi–Galego prove the dimension- $|G|$ upper bound by putting the regular representation of G on $\ell_2(G)$ and multiplying everything by the central sign. One can remove the constant vector instead. On the augmentation hyperplane

$$V = \left\{ (a_h)_{h \in G} \in \ell_2(G) : \sum_{h \in G} a_h = 0 \right\},$$

the regular action is still faithful, and, for $|G| > 2$, no non-trivial group element acts as $-\text{Id}_V$. Therefore the extra sign in $\{-1, 1\} \times G$ can act by the scalar -1 on V and on the complement of V in X , giving a faithful finite group of isomorphisms containing the global $-\text{Id}_X$. The finite renorming theorem of Ferenczi–Galego then turns this finite group into the full isometry group for an equivalent norm.

Lemma 1 (Augmentation representation avoids $-\text{Id}$). *Let G be a finite group of order $n \geq 3$. Let λ be the left regular representation of G on $\ell_2(G)$ and let*

$$V = \left\{ (a_h)_{h \in G} : \sum_{h \in G} a_h = 0 \right\}.$$

Then the restricted representation $\pi(g) = \lambda(g)|_V$ is faithful and $\pi(g) \neq -\text{Id}_V$ for every $g \in G$.

Proof. The subspace V is invariant under all permutations of the basis, hence under $\lambda(G)$. Let

$$u = \frac{1}{n} \sum_{h \in G} e_h.$$

If $\pi(g) = \text{Id}_V$, then $\lambda(g)$ fixes V and also fixes u , so it fixes $\mathbb{R}u \oplus V = \ell_2(G)$. Thus $\lambda(g) = \text{Id}$ and $g = e$.

Now suppose, toward a contradiction, that $\pi(g) = -\text{Id}_V$. For each $h \in G$, the vector $e_h - u$ belongs to V , so

$$e_{gh} - u = \lambda(g)(e_h - u) = -(e_h - u) = -e_h + u.$$

Equivalently,

$$e_{gh} + e_h = 2u = \frac{2}{n} \sum_{k \in G} e_k.$$

Since $n \geq 3$, choose $k \in G$ distinct from both h and gh if $gh \neq h$; the coefficient of e_k is then 0 on the left and $2/n$ on the right, a contradiction. If $gh = h$, then $g = e$, which gives $\pi(g) = \text{Id}_V$, not $-\text{Id}_V$. Hence no g acts as $-\text{Id}_V$. $\square$

Theorem 1. *Let G be a finite group of order $n \geq 3$. If X is a separable real Banach space with $\dim X \geq n - 1$, then $\{-1, 1\} \times G$ is representable in X .*

Proof. Let V and π be as in Lemma 1. Since $\dim V = n - 1$ and $\dim X \geq n - 1$, choose a closed complement Y so that, as a vector space and up to linear isomorphism,

$$X \simeq V \oplus Y.$$

Equip this decomposition temporarily with the Hilbertian sum norm

$$\|(v, y)\|_0 = (\|v\|_2^2 + \|y\|_Y^2)^{1/2},$$

where $\|\cdot\|_Y$ is any Banach norm on Y induced by the chosen isomorphism. This is an equivalent norm on X .

For $(\varepsilon, g) \in \{-1, 1\} \times G$, define

$$A_{\varepsilon, g}(v, y) = (\varepsilon \pi(g)v, \varepsilon y).$$

These operators form a finite group of isometries of $(X, \|\cdot\|_0)$ and $A_{-1, e} = -\text{Id}_X$. The representation is faithful. Indeed, if $A_{\varepsilon, g} = \text{Id}$, then on V one has $\varepsilon \pi(g) = \text{Id}_V$. If $\varepsilon = 1$, Lemma 1 gives $g = e$. If $\varepsilon = -1$, then $\pi(g) = -\text{Id}_V$, which is impossible by the same lemma.

Thus $\{-1, 1\} \times G$ is isomorphic to a finite group of isomorphisms of X containing $-\text{Id}_X$. By Ferenczi–Galego’s finite renorming theorem [1, Theorem 8], equivalently Theorem 8 in their arXiv numbering, there is an equivalent norm on X for which this finite group is exactly $\text{Isom}(X)$. Hence $\{-1, 1\} \times G$ is representable in X . $\square$

Corollary 1 (Negative answer to Question 13). *There are no arbitrarily large integers n with the property asked for in Question 13 of [1]. More strongly, for every finite group G of order $n \geq 3$, $\{-1, 1\} \times G$ is representable in every separable real Banach space of dimension at least $n - 1$.*

Proof. If a group G of order $n \geq 3$ had the property in Question 13, then $\{-1, 1\} \times G$ would fail to be representable in any separable real Banach space of dimension $< n$. Taking any separable real space of dimension $n - 1$ contradicts Theorem 1. $\square$

Verification and Edge Cases

The proof uses only two ingredients: the elementary augmentation representation lemma and the finite renorming theorem from [1]. The case $n = 2$ is intentionally excluded: if $G \simeq \mathbb{Z}/2\mathbb{Z}$, the augmentation representation is one-dimensional and the non-trivial element acts as $-\text{Id}$, so the signed product representation would not be faithful in dimension 1. This exception does not affect Question 13, which asks about arbitrarily large n .

No computational verification was needed. The main human-review points are:

- the assertion that any finite-dimensional subspace of a Banach space has a closed complement, used to write $X \simeq V \oplus Y$;
- the faithfulness check when $Y = 0$, where the condition $\pi(g) \neq -\text{Id}_V$ is essential;
- the invocation of [1, Theorem 8], which applies to finite groups of isomorphisms of a separable real Banach space containing $-\text{Id}$.

Novelty Check

Local run indexes were searched for 0706.3861, the paper title, “representable in a Banach space”, “normal subgroup with two elements”, and the finite-dimensional question; no prior packet for this dimension improvement was found. Web searches on July 3, 2026 for phrases including “ $\{-1, 1\} \times G$ $|G|-1$ Banach displayable”, “augmentation representation Jarosz Banach space isometries”, and “For arbitrarily large n representable separable real Banach” did not locate an existing statement of this $n - 1$ bound. A later survey/source paper by Ferenczi–Rosendal [2] was also checked; it recalls the Ferenczi–Galego finite theorem but does not appear to record this augmentation-hyperplane sharpening. Novelty confidence is therefore moderate: the proof is elementary and may be folklore, but it is not already recorded in the run indexes or the bounded search performed for this packet.

References

References

- [1] V. Ferenczi and E. M. Galego, *Countable groups of isometries on Banach spaces*, Trans. Amer. Math. Soc. 362 (2010), no. 8, 4385–4431. arXiv:0706.3861.

- [2] V. Ferenczi and C. Rosendal, *Displaying Polish groups on separable Banach spaces*, arXiv:1110.2970.